Developing New Tools to Search for The Most Extreme Outflows in the Universe Using Python and Machine Learning

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Quasars are among the most luminous active galactic nuclei (AGN) powered by massive, luminous accretion disks surrounding supermassive black holes at the center of most galaxies. Launching from these accretion disks are energetic outflows of gas and radiation. Among these, quasar extremely high velocity outflows (EHVOs) are some of the fastest known outflows, consisting of gas observed at speeds faster than 10% the speed of light. In the past few years, over 150 EHVO cases have been confirmed. Their prevalence across AGN populations suggests EHVOs are not rare anomalies but a common feature of quasar activity in the early universe. The massive kinetic power released by EHVOs is suspected to have a critical influence on early galaxy formation and evolution including black hole growth and star formation. However, the fastest EHVOs with speeds greater than 20% the speed of light remain underrepresented in existing surveys due to the detection limits of conventional spectra analysis methods. To date, only two have been discovered. We present an automated detection pipeline to identify EHVOs with velocities exceeding those measured by most of the previous studies. The basis of our procedure uses multi-epoch ratios of flux observations to suppress relatively non-variable Lyman-Alpha forest absorption and isolate fluctuating absorption intrinsic to the quasar environment. Using this pipeline, we have demonstrated a successful identification for one of the two known quasars that exhibit an EHVO faster than $0.2c$. To improve detection accuracy, we additionally explore the integration of SpenderQ, a redshift-invariant autoencoder, as a machine learning tool for reconstructing the unabsorbed quasar continuum directly from the observed spectrum. This approach shows promising initial results as a path toward detection in $v > 0.2c$ cases where multi-epoch ratios alone cannot isolate the distinctive absorption features.

CONTENTS

1. INTRODUCTION

Quasars are powered by supermassive black holes (SMBH). As surrounding gas falls inward towards the SMBH, it forms a massive, rotating accretion disk that releases enormous amounts of energy through gravitational compression and friction, making quasars detectable across much of the observable universe [13]. The intense radiation field ionizes surrounding gas, producing the characteristic broad emission lines visible in quasar spectra [14]. This energetic environment drives powerful outflows of ionized gas, including EHVOs ejected at speeds exceeding 10% the speed of light ($v > 0.1c$) [21]. Understanding these outflows is critical, as their kinetic energy output is thought to regulate star formation and black hole growth across cosmic time [12].

As light from high-redshift quasars travels to us from the early universe, it encounters a myriad of hydrogen gas clouds in the intergalactic medium between galaxies. Quasar EHVOs show broad, significant absorption, particularly in CIV, SiIV, and NV, and shift these lines into the Lyman-Alpha (Ly-α) region [20]. Every hydrogen cloud the light passes through has a smaller redshift than the quasar itself, causing absorption lines to appear at different wavelengths blueshifted (shorter) than the quasar's own emission line, resulting in the dense ”forest” of absorption lines. This collection of small lines obscures the characteristic absorption features we search for to identify EHVOs.

A new higher velocity range of outflows recently discovered with speeds greater than 20% the speed of light are blueshifted into this region, making it difficult to distinguish their intrinsic quasar background from the intergalactic medium [9]. Relative to the timescales in which we observe variability of a quasar's intrinsic absorption [1], the Ly-α lines remain relatively constant due to the non-variability of the hydrogen gas clouds [17]. By dividing multiple spectral observations of a potential EHVO, the Ly-α lines are suppressed to the baseline leaving behind isolated lines that can then be analyzed to determine the class of outflow from the quasar's intrinsic absorption profile.

2. SCIENTIFIC BACKGROUND

2.1 Quasar Spectroscopy

Astrophysical spectroscopy relies on the evaluation of a quasar's spectral energy distribution, typically formatted as flux density against wavelength. Quasars emit large amounts of energy across the electromagnetic spectrum, characterized by a smooth continuum of light emitted by the accretion disk, layered with emission lines from photoionized gas winds [14]. As the universe expands, the space between Earth and all celestial objects including quasars constantly increases

[16]. Since these active galactic nuclei (AGN) are located at immense cosmological distances, the expansion of the universe significantly stretches the light traveling toward us, shifting these emission profiles to longer wavelengths on the electromagnetic spectrum [16]. This cosmological redshift (z) is defined by the following relation:

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} \tag{1}$$

where $\lambda_{\text{observed}}$ represents the wavelength recorded in our observer frame of reference on Earth and λ_{rest} denotes the laboratory rest-frame wavelength of the atomic transition.

In contrast to the uniform cosmological redshift affecting the entire quasar, intrinsic outflows of gas travel directly along our line of sight toward observational telescopes. This high-velocity spatial motion introduces a local, kinematics-driven Doppler blueshift. Consequently, the intrinsic absorption features created by these outflows appear at shorter (blueshifted) wavelengths relative to the quasar's emission redshift (z_{em}).

2.2 Spectral Characteristics of Extremely High-Velocity Outflows

Quasar outflows are identified, among other ways, via broad absorption lines (BALs), which emerge when highly ionized streams of gas and plasma intercept the background quasar continuum along our line of sight and produce an absorption profile wider than 2,000 km/s. The ions used to track these high-ionization outflows are resonance doublets, typically Carbon IV (CIV $\lambda 1548.20, 1550.77$ Å), Silicon IV (SiIV $\lambda 1393.76, 1402.77$ Å), and Nitrogen V (NV $\lambda 1238.82, 1242.80$ Å) [19] In the $v > 0.2c$ we are now exploring, only CIV is expected to be identifiable among the deeper region of the Ly-α forest (see Section 2.3).

Extremely High-Velocity Outflows (EHVOs) represent the most extreme end of this phenomenon, defined as outflows often observed at velocities between 10% and 20% of the speed of light ($0.1c < v < 0.2c$). More than 150 cases have been confirmed in recent surveys within this velocity range (Candelaria-Stoner 2026, in preparation) [19]. The interest in EHVOs comes from their energy output: the kinetic power launched into the surrounding interstellar environment scales proportionally to the velocity cubed ($E \propto v^3$). This makes EHVOs candidate drivers of large-scale active galactic nucleus (AGN) feedback, a process thought to regulate star formation, clear gas reservoirs, and power the co-evolution of host galaxies and their central supermassive black holes [18] [12] [4].

Numerical simulations by Di Matteo et al. (2005) prove that a central black hole must actively push energy back into its host galaxy to stop runaway growth [4]. As a black hole rapidly accretes material, it releases energy back into the surrounding medium. Their model establishes a parameter: Approximately 5% of this radiated energy couples thermodynamically with the nearby interstellar gas, generating a powerful kinetic wind. The kinetic power of the wind is strong enough to expel the remaining gas from the galaxy center. This expulsion of gas stops further black hole growth entirely, causing the black hole mass to saturate, and the sudden lack of cold gas shuts down all star formation activity. Without this kinetic energy clearing the gas, the final host structure would retain massive reservoirs of cold material and continue forming stars indefinitely. Calculating the actual kinetic energy output (E_{k}) of a quasar outflow into its host galaxy depends fundamentally on the velocity cubed (v^3) [19]. Considering how power scales with this cubic relationship, a minor increase in speed causes a massive surge in energy. This means an outflow moving at 0.2c carries roughly one to two and a half orders of magnitude more energy than one moving at the high-velocity threshold of 5,000–10,000 km s^{-1}, assuming similar distances from the central source [19]. An outflow traveling at $0.2c$ delivers eight times more kinetic power than a more common wind moving at $0.1c$. Average quasar winds with $v > 0.1c$ barely touch the 5% threshold needed for feedback, however the extreme velocity of an EHVO at $v > 0.2c$ may meet or even exceed this limit. These outflows carry more than enough energy to dominate the evolution of a galaxy's structure. AGN feedback is depended on in cosmological simulations to explain why massive galaxies stop forming stars, why the mass of a galaxy's central black hole correlates so tightly with the mass of its stellar bulge, and how the large-scale structure of the universe is sorted [5]. The current literature is based on assumptions necessitated by a lack of evidence for feedback mechanisms with greater kinetic power. If the fastest outflows are significantly more prevalent than previously estimated, the energy launched into the IGM at high redshift ($z \approx 4 - 5$) soon may need to be revisited to include effects on the predicted rates of star formation suppression and the mass correlation of supermassive black holes and their host galaxies.

2.3 The Lyman-Alpha Forest and the Intergalactic Medium

As light from a high-redshift quasar ($z_{\mathrm{em}} \gtrsim 2$) travels to our telescopes from the early universe, it encounters a vast number of hydrogen gas clouds distributed across the intergalactic medium (IGM) between galaxies. Each foreground cloud contains neutral hydrogen (HI) that absorbs photons matching the resonant Lyman-alpha electron transition [16]. This specific atomic transition occurs when an electron jumps from the ground state ($n = 1$) to the first excited state ($n = 2$), corresponding to a laboratory rest wavelength of 1215.67 Å [16].

These intergalactic clouds exist at varying distances along our line of sight, meaning each individual cloud possesses a smaller redshift than the quasar itself ($z_{\rm abs} < z_{\rm em}$). Each cloud leaves a narrow absorption line at a unique observed wavelength:

$$\lambda_{\rm observed} = 1215.67 \text{ Å} \times (1 + z_{\rm abs}) \tag{2}$$

The combined effect of light passing through multiple clouds at different degrees of redshift is a "forest" of narrow absorption lines located blueward of the quasar's own Ly-α emission lines (Figure 1). This dense collection of narrow lines interferes with the spectral baseline, making it exceptionally difficult to map the underlying quasar continuum and obscuring the broader absorption features that signal intrinsic phenomena such as extremely high velocity outflows [16].

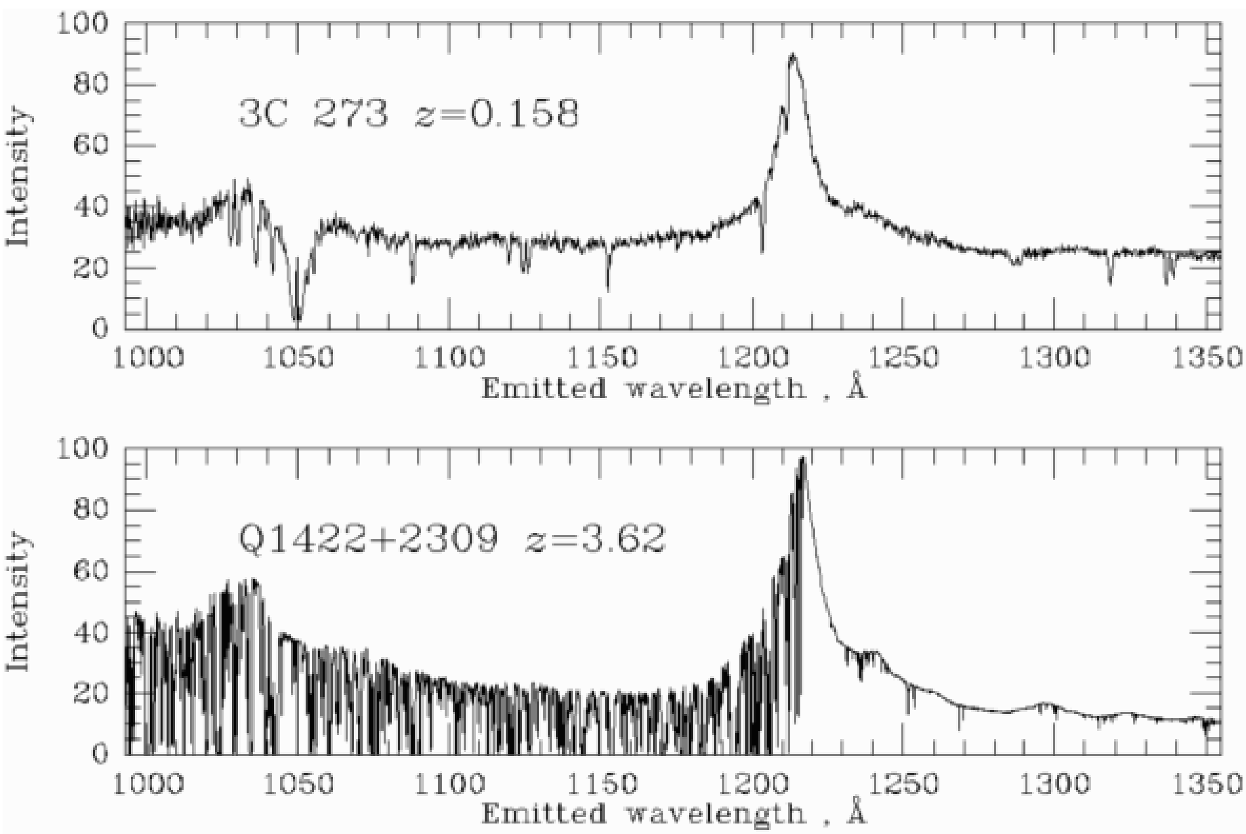

FIG. 1. The Lyman-alpha forest (b), a series of absorption lines produced by hydrogen gas clouds in the intergalactic medium that lies between us and our line of sight of a quasar. Sharp, narrow absorption caused by hydrogen gas clouds in the IGM obscure absorption toughs blueward of 1215.67 Å. Credit: UCLA Division of Astronomy & Astrophysics.

2.4 The Unique Challenge of the $v > 0.2c$ Domain

The catalogue of EHVOs up to $0.2c$ is increasingly updated, however outflows with speeds exceeding 20% of the speed of light ($v > 0.2c$) remain severely underrepresented in existing surveys, with only two documented to date [22] [8]. This does not imply a physical upper limit on outflow mechanics; rather, it reflects the detection limits of conventional spectral analysis methods.

When an outflow moves faster than $0.2c$, its CIV absorption feature is blueshifted, pushing it entirely out of the clean continuum redward of the Ly-α line and placing it directly into the highly obscured Ly-α forest region [22]. In this region of the observed quasar spectrum, the broad, intrinsic absorption trough from the outflow completely blends with the dense grouping of narrow absorption lines caused by hydrogen gas in the IGM.

Conventional automated continuum-fitting techniques and visual inspections fail in this upper velocity limit. Standard analysis algorithms misinterpret the wide, continuous dip of a potential outflow in this range as a series of standalone variations in flux, leading to underestimation or total omission of EHVO absorption profiles. Searching this region requires tools designed to isolate the non-variable Ly-α lines from the active quasar environment. Intergalactic gas clouds remain stable over millions of years relative to intrinsic EHVO absorption lines which frequently vary on timescales of months to years [1]. Based on this context, spectral ratio comparisons provide a viable method to suppress the non-variable 'Ly-α forest and leave behind isolated, ultra-high-velocity absorption features for analysis. The following section will outline the entire process used in this research to identify absorption. For clarity on terminology, a full list of defined terms has been included in the Glossary of Terms at the end of this paper.

3. METHODOLOGY

The detection pipeline is a modular, Python-based architecture designed to process a large volume of Sloan Digital Sky Survey (SDSS) quasar spectra observations over a directory of EHVO candidates simultaneously. The process is structured in three stages: normalization, spectral ratio analysis, and ion absorption identification.

Stage 1: Normalization

Analyzing spectra typically requires a continuum baseline against which absorption features can be measured. Quasar spectra come from SDSS in the format of flux density as a function of observed wavelength, but are not normalized. Our normalization module uses continuum fitting to divide out the underlying power-law continuum of the quasar, which sets the spectral flux to a normalized baseline. This ensures that absorption troughs are expressed as fractional deviations from the continuum rather than as absolute flux values. This format is necessary for consistent comparison across quasars at different redshifts and luminosities.

Stage 2: Isolating Features via Multi-Epoch Spectral Ratios

For outflows exceeding $0.2c$, the CIV absorption trough is often pushed into the Ly-α forest region of the observed spectrum, where it is blended with the dense forest of narrow Ly-α lines. Standard continuum-fitting and visual inspection methods fail in this upper velocity range, as described in Section 2. Dividing multiple spectral observations addresses this by taking advantage of physical differences between the two overlapping features.

Hydrogen gas clouds responsible for the Ly-α forest lines are stable over cosmological timescales of millions of years and do not vary significantly between observations. The intrinsic absorption features of EHVO winds, conversely, are known to vary on timescales of days to months, driven by changes in the ionization state or covering fraction of the outflowing gas [1]. This difference in variability timescales is the physical basis of this method.

By obtaining multiple spectral observations of the same quasar from different dates (epochs) and computing their ratio (epoch A / epoch B), the non-variable Ly-α forest lines divide to 1 and are suppressed close to the baseline. Any spectral feature that has varied between observation dates—including intrinsic EHVO absorption—remains as residual flux in the ratio plot. Spectral division was chosen over subtraction because it normalizes for differences in the overall flux level between observations. This creates a dimensionless ratio that isolates absorption variability between observations.

Candidate EHVOs are confirmed by inspecting consistent EHVO absorption features in the same wavelength regions across multiple independent ratio plots created from different pairs of divided observations. Each ratio consists of the same reference observation in the numerator. Across all observations, the spectrum that shows the strongest absorption of NV or CIV is selected as the reference to be divided by all other observations individually. A feature that persists across several ratio combinations is unlikely to be instrumental noise or an anomaly of a single observation, which increases our confidence that the absorption is truly intrinsic to the outflow.

Stage 3: Ion Absorption Identification

Once the continuum is established, the pipeline searches for the characteristic absorption profiles produced by EHVOs. One of the primary identifiers used is the CIV ion. In rest frame, it appears near $\approx 1550\mathring{A}$ but is blueshifted to shorter wavelengths as outflow velocity increases. For outflows exceeding $0.1c$, blueshifted CIV absorption is searched for across the corresponding wavelength range in the observed spectrum.

The output ratio spectra from Stage 3 are passed to the automated absorption detection module, which analyzes each ratio plot for statistically significant absorption troughs in the wavelength range corresponding to CIV velocities greater than $0.2c$. The code flags regions where the 1. ratio flux drops below 10% of the continuum over a sufficiently broad wavelength interval, and 2. the width is greater than 2000 km/s. These requirements are consistent with the expected width of a broad absorption feature at these velocities. The pipeline is designed to run over an entire directory of observations and generate flagged absorption output plots for each quasar. This pipeline will allow for re-examination of SDSS cases from past studies [15] in which velocities above $0.2c$ were not previously searched for.

4. RESULTS: EHVO WITH $V > 0.2c$ DETECTED

To validate the pipeline, we first applied the full detection process to a confirmed EHVO quasar, J2318. This quasar was selected as it is one of the only two quasars currently confirmed to have an EHVO with a velocity greater than $0.2c$ [22] [8].

After normalization, the ratios were generated through the division module by dividing the three existing spectral observations of the quasar against one another. The resulting ratio plots reveal a deep, broad absorption trough spanning approximately 1140–1190 Å in the observed frame. For reference, the CIV doublet appears near $\approx 1550\ \mathring{A}$ in the rest frame. The degree of blueshift evident in the ratio spectrum corresponds to an outflow velocity exceeding $0.2c$, placing this candidate among the fastest known quasar outflows identified to date. This result aligns with the confirmed velocity of J2318 ($v \approx 0.3c$) [22], verifying that our pipeline can systematically identify EHVOS in this higher velocity range.

The continuous shape of the absorption trough spanning roughly 50 Å (Figure 2) is not consistent with the narrow, isolated features characteristic of Ly-α forest lines. Its width and depth are consistent with the absorption line profiles expected from a quasar EHVO. The Ly-α forest

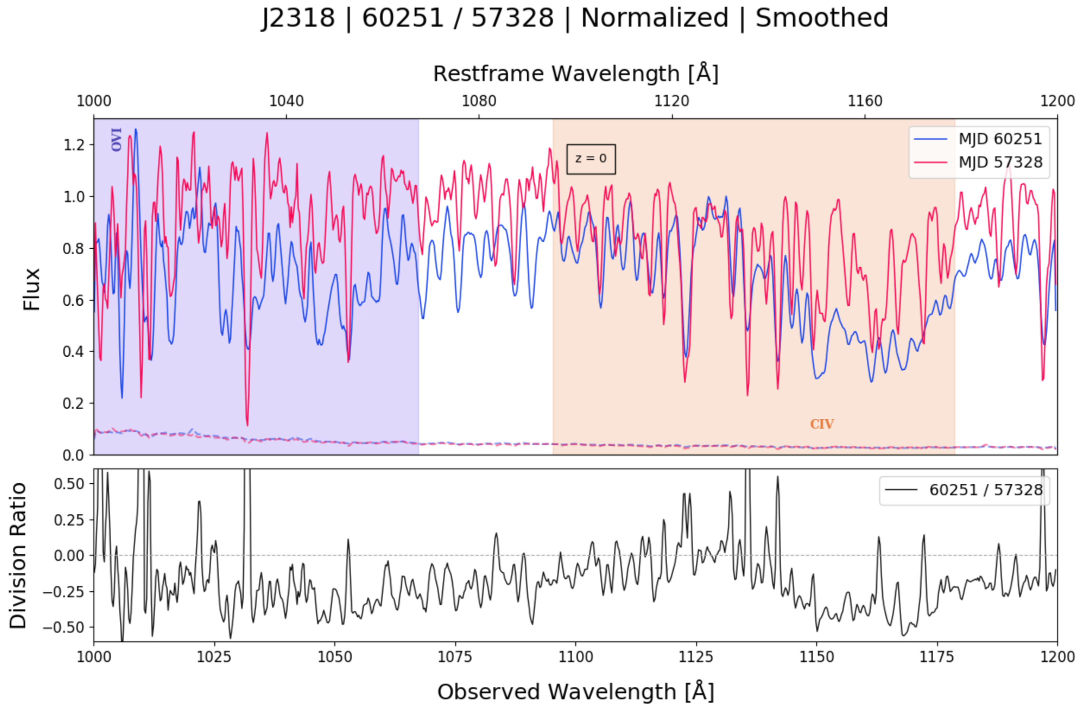

FIG. 2. Two observations of quasar J2318 and their ratio. By dividing two observations at different epochs, we suppress absorption caused by the Lyman-alpha region to highlight intrinsic flux changes. In the ratio (lower panel) the Lyman-alpha lines have been suppressed to the baseline, allowing the EHVO features to be isolated. CIV absorption is evident in the highlighted region from approximately 1135 Å to 1190 Å, indicating blueshifted absorption intrinsic to the EHVO. This feature establishes quasar J2318 as one of only two known quasars with an EHVO faster than 0.2c.

lines in the ratio plot are substantially suppressed toward the baseline, which confirms that the division method successfully isolated the variable intrinsic absorption from the non-variable IGM background.

To assess detection confidence, ratio plots were constructed from two independent pairs of observations available for J2318. The consistency of the absorption signature across both available ratio combinations provides strong evidence that the feature is intrinsic to the quasar outflow rather than a result of the Ly-α forest, unrelated absorption or observational differences.

The ratio spectra for J2318 were then passed through our automated absorption detection module described in Section 3. The module flagged the absorption trough at 1140–1190 Å as a statistically significant absorption feature, consistent with the manual identification from the generated ratio plots (Figure 2). This agreement between the automated output and the visual analysis of the ratio spectra confirms that the pipeline performs reliably in this velocity domain and does not require manual inspection to isolate the feature from the Ly-α forest.

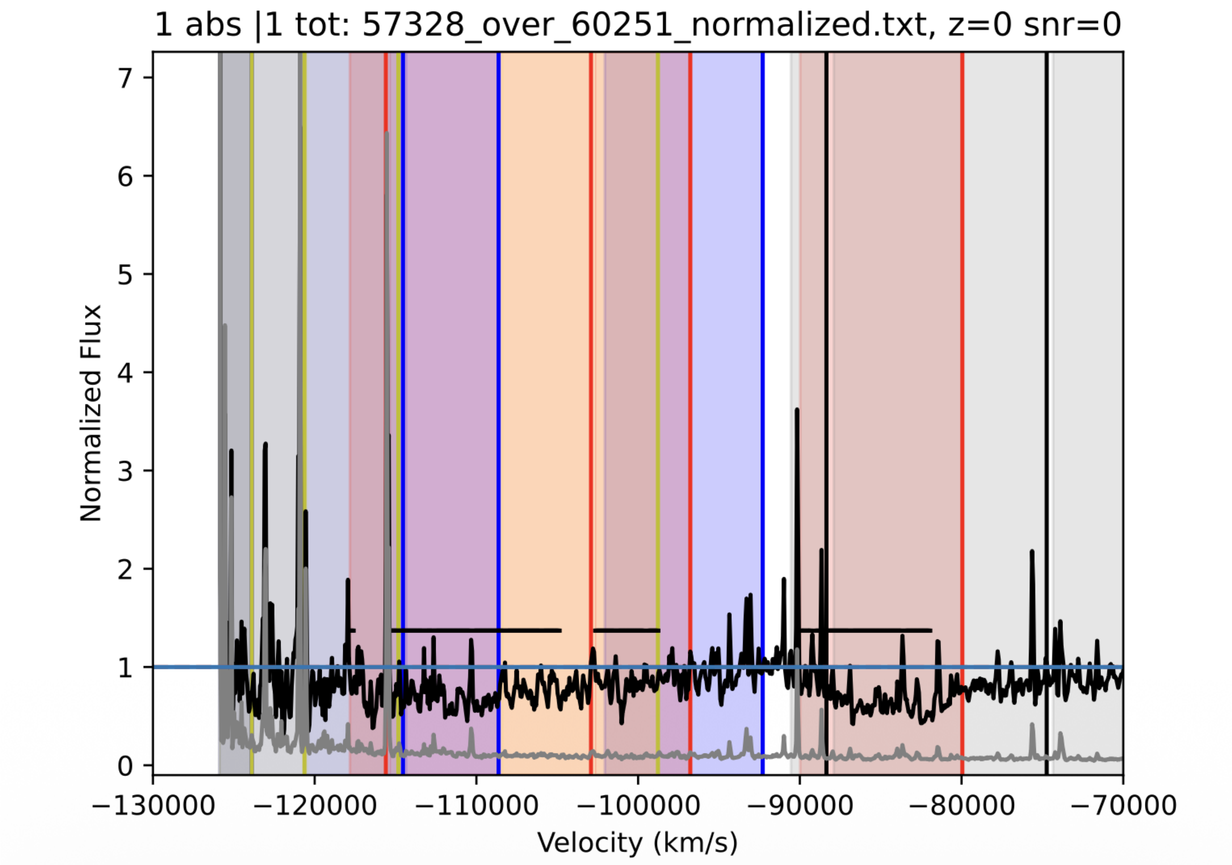

FIG. 3. The ratio of two observations of quasar J2318 at epochs MJD 57328 and MJD 60251, analyzed by our automated absorption detection module. Absorption 10% below the continuum and $> 10km/s$ is flagged by the program, indicated by the horizontal black line spanning -83,000 km/s to -90,000 km/s. This trough aligns with the velocity range measured by Rodriguez Hidalgo et al., 2026 [22].

The results for J2318 show multiple lines of evidence for a confirmed EHVO at $v > 0.2c$: a broad CIV absorption trough blueshifted far into the Ly-α forest region, consistent absorption features across two independent observation ratios, and confirmation from our automated detection module(Figure 3). These results establish first evidence that our pipeline successfully systemizes identification in the velocity domain where conventional spectral analysis methods either fail or require extensive manual fitting. It provides a reproducible framework for extending the search to a larger catalog of SDSS quasar candidates.

To verify that this pipeline is suited to detect blueshifted CIV in outflows with $v > 0.2c$, we additionally selected five confirmed EHVO cases from DR16 that showed strong NV absorption based on visual inspection from prior variability studies [15]: J103800.50 ($z = 2.702$), J123830.71 ($z = 2.846$), J112954.72 ($z = 3.07$), J103119.22 ($z = 3.194$), and J230820.39 ($z = 3.623$). Since NV is heavily blueshifted into the Ly-α forest region for $v > 0.1c$, we searched these spectra for NV absorption in the velocity range corresponding to expected CIV in $v > 0.2c$ spectra. Since CIV absorption will be blueshifted into the Ly-α forest region for EHVOs with $v > 0.2c$, successfully detecting NV absorption in the the Ly-α forest region will validate this approach for searching for CIV absorption in future studies.

5. MACHINE LEARNING APPROACH TO EHVO FEATURE ISOLATION

5.1 Limitations of Conventional EHVO Feature Extraction Methods

The spectral ratio method described in Section 3 represents a tool for isolating the intrinsically variable absorption features of EHVO candidates from the non-variable Ly-α forest. However, its effectiveness depends on a physical assumption: that the absorption features of the EHVO must vary between the two epochs of observation being divided. This assumption holds true in many cases, since EHVO outflow absorption is known to vary on timescales of days to months [1].

Still, there exists a substantial subset of EHVO candidates for which the absorption remains approximately constant across different epochs of observation, causing the spectral ratio to converge near 1 and leaving the absorption feature invisible. As we expand our search for quasars with EHVO velocities exceeding $0.2c$, we suspect that outflows with similar absorption will persist among this upper velocity range of EHVOs. Standard approaches to normalizing the spectra before division rely on assumptions about the physical nature blueward of the Ly-α line, which introduces further uncertainty in the true flux values. These limitations motivate a novel approach: training an autoencoder to learn the intrinsic, unabsorbed quasar continua from raw data, normalizing spectra via division by these unabsorbed continua, then using the ratio method to isolate highly variable EHVO absorption features using a more accurately-modeled baseline.

Previous approaches to modeling intrinsic quasar continua have included power-law fits to continuum windows free of strong emission or absorption (as used in [19]) and parametric fits to emission-line profiles. While these approaches are effective for specific spectra in the ($v < 0.2c$) region and work well for the normalization step of the existing pipeline, they require prior knowledge about which spectral regions are free of absorption, which remains impossible deep in the Ly-α forest.

The true unabsorbed quasar continuum, $C_{q\lambda}$, is not directly observable. Instead, what is measured is the flux $f_{q\lambda}$, which is a product of the intrinsic continuum, the mean intergalactic absorption $\bar{F}(z_\lambda)$, and the fluctuating transmission field $\delta^t_{q\lambda}$:

$$f_{q\lambda} = C_{q\lambda}\,\bar{F}(z_\lambda)\,(1 + \delta^t_{q\lambda})$$

Recovering intrinsic absorption requires dividing the observed flux by an estimate of $C_{q\lambda}\,\bar{F}(z_\lambda)$. Standard pipelines such as PICCA estimate this term by fitting each spectrum to a universal quasar template scaled by individual quasar parameters that absorb variations in luminosity and spectral index [2]. This process introduces a systematic distortion that has been well-documented.

Busca et al. (2025) demonstrate that this continuum fitting mixes absorption information across wavelengths within a single forest. The fitted parameters for a quasar (q) absorb a weighted mean of the true fluctuations across all pixels in the forest, meaning the measured field $\delta^m_{q\lambda}$ is not equal to the true field and instead takes the form:

$$\delta^m_{q\lambda} \approx \delta^t_{q\lambda} - \epsilon_{q\lambda} + \text{higher-order terms}$$

where $\epsilon_{q\lambda}$ is the mean intra-forest fluctuation weighted by a wavelength-dependent factor. As a result, the correlation signal at any given separation receives contaminating contributions from pixel pairs at very different separations. The resulting bias in inter-forest correlations is a structured distortion that must be modeled and corrected, typically through the distortion matrix. This correction factor introduces biased error which can cause non-physical features to be included in the fitted continuum, especially blueward of the Ly-α line where absorption profiles are difficult to observe. This is a known limitation of PCA analysis that has been well documented [3]. This motivates the use of an autoencoder trained to reconstruct the intrinsic quasar continuum without reference to the forest or any intervening absorption. By training the model to learn only the smooth spectral shape defined by the unabsorbed quasar emission, the reconstruction serves as a physically-informed baseline that suppresses the distortion described by Busca et al., 2025. As they describe, a sufficiently rigid continuum estimate drives the residual $\epsilon_{q\lambda} \to 0$, reducing the absorption contamination in the modelled continuum. The distortion matrix correction becomes unnecessary to the degree that the continuum estimator itself can be constrained not to absorb the region of potential EHVO absorption.

5.2 Autoencoders

An autoencoder is a class of neural network (a type of machine learning (ML) model) designed for unsupervised learning. The network consists of two components: an encoder, which maps the high-dimensional input to a compressed, low-dimensional representation called the latent space, and a decoder, which reconstructs the original input from the latent representation [6]. The program is forced to represent all relevant information in a lower-dimensional format, so training an autoencoder on a large dataset encourages the network to learn only the most statistically dominant modes of variation in the data [7]. In this context, the input is the full flux array across the observed wavelength range (a vector of several thousand flux values) and the output is a reconstruction of that same spectrum. The latent representation, which consists of only ten parameters, can be thought of as capturing the main physical properties of the quasar: for example, the slope of the power-law continuum, the strength and width of the CIV emission line, and the strength of NV emission [7].

5.3 SpenderQ Architecture and Advantages

The specific autoencoder model chosen for this work is SpenderQ, a neural network architecture developed for dimensionality reduction and reconstruction of unabsorbed quasar spectra[7]. SpenderQ was developed with several properties that make it uniquely well-suited for the problem of EHVO detection in the Ly-α forest. The most useful of these properties is redshift invariance. Since quasars in the SDSS are observed at a wide range of emission redshifts (for our sample, $4 < z_{\rm em} \gtrsim< 5$, a model trained on spectra at one redshift would not generalize to spectra at another if it operated in observed wavelength space. SpenderQ solves this by operating in the quasar rest frame. Spectra are shifted to rest-frame wavelength before being passed to the network, so the model learns a redshift-independent representation of quasar spectra structure. This eliminates the need to train separate models for different redshift ranges, which would be inefficient given the finite size of training data available for EHVO candidates. A second key property is that SpenderQ, as an autoencoder, is trained on the raw observed data without requiring separate mock spectra or clean training sets. Conventional supervised approaches to continuum modeling require a set of spectra with known, labeled continua to serve as the training data. These labels are difficult to produce for quasar spectra, since the true unabsorbed continuum is never directly observed. SpenderQ surpasses this requirement by learning a low-dimensional representation of the raw data distribution. The network is not told what the continuum should look like for any individual spectrum, but rather learns what is common across the full range of spectra [11]. The third advantage is that SpenderQ was trained in the rest-frame wavelength range of $800 - 3170\mathring{A}$, aligning with the rest-frame wavelength range of $1000 - 1200$ where we are searching for CIV absorption. The network encodes each input spectrum into approximately 10 latent parameters. These parameters are not labeled in advance but are expected to correspond, following training, to physically meaningful spectral properties: for example the strength of ion emission lines, the slope of the power-law continuum, and the overall normalization. The decoder then maps these parameters back to a full reconstructed spectrum. In a recent report by Hahn et al. [7], the reconstructions showed high accuracy within $< 5\%$ in the Ly-α forest region.

5.4 Dataset

For testing SpenderQ, raw SDSS quasar spectra as SDSS .FITS files were selected from the same parent sample as the ratio method described in Section 3. An autoencoder does not require pre-labeled continua, meaning the training set is directly from the SDSS spectral archive without additional changes beyond the redshift and signal-to-noise requirements from the existing modules. This is another advantage, as the number of confirmed EHVO cases remains small relative to the full quasar population. A supervised approach would require the confirmed EHVOs to be withheld from training as a separate test set, which would further limit the available data. Before being fed to the network, each spectrum will be shifted to its rest frame using the emission redshift provided by the survey and interpolated onto a common rest-frame wavelength grid. Next, the flux is normalized by the median flux in the rest-frame $1445 - 1455\mathring{A}$ and $700 - 1705\mathring{A}$ continuum windows.

5.5 Reconstructing the Unabsorbed Quasar Continuum

The purpose of using an autoencoder within the detection pipeline is to accurately reconstruct the unabsorbed quasar continuum above the dense Ly-α forest region without depending on assumptions or arbitrary non-physical parameters. The autoencoder learns the common parameters of variation among quasar continua and emission features, and treats the Ly-α forest as a high-frequency pattern that varies from quasar to quasar and is not efficiently encoded in the low-dimensional latent space [11]. The narrow and broad absorption features of the forest are masked out of the reconstruction. This property is the same physical one exploited by the ratio method, but through an entirely different tool. The autoencoder reconstructs the continuum by learning what is common across thousands of quasar spectra then encodes only that shared structure. The ratio method alone works best when the EHVO absorption is highly variable. In cases where EHVO absorption does not vary significantly across observations, the variability in flux is weak but may still be recoverable. Dividing each spectra by the autoencoder-reconstructed unabsorbed continuum before dividing the multi-epoch spectra further amplifies flux changes caused by fast-moving EHVOs. After reconstruction, the flux variation between the observed spectrum and the autoencoder reconstruction can be analyzed for absorption features. If EHVO absorption is present, it should appear as a broad absorption trough in this ratio at the expected wavelengths corresponding to blueshifted ions (CIV or NV) in the quasar rest frame.

5.6 Limitations of the Autoencoder Approach and Adjustments

Autoencoder-based continuum reconstructions are limited by their dependence on the specific distribution of quasar spectra used as training data. This can potentially result in the reconstructed continuum being too low or too high near the absorption region. If it is too low, true absorption can look weaker or disappear. If it is too high, slight spectral variation unrelated to EHVOs can look like artificial absorption. Since EHVOs with $v < 0.2c$ are rare, they are unlikely to be strongly represented in the training set and learned as typical continuum structure. This reduces the risk that SpenderQ will absorb the EHVO feature into the reconstructed continuum as one of ten common parameters. However, biases towards $v < 0.2c$ continua may remain if quasars containing EHVOs in this velocity range have continuum or emission properties that differ significantly from the training set. Several adjustments are being considered to model the true continuum more accurately. The first is absorption masking: the sigma variable in the reconstruction layers of the underlying Spender encoder can be lowered to mask wider spectral features on the scale of typical EHVO absorption trough widths (¿ 1000 km/s [21] before they enter the latent representation [11] to ensure that broad absorption troughs are not encoded and do not appear in the reconstruction. The second considered adjustment is reducing latent dimensionality. Reducing the number of latent parameters further limits what the network can encode into the reconstructed continuum. Using fewer parameters ensures the network is forced to represent only the most common spectral properties (continuum slope, overall emission line strengths) and cannot encode the detailed structure of individual absorption features [11]. Doing so will require retraining the model on a new set of spectral data, such as the DESI mocks originally used by Hahn et al. for training [7]. The trade-off is that too few parameters will result in poor reconstruction of the quasar continuum itself, especially in the Ly-α forest in the range of $v > 0.2c$. An alternative approach is to train the network on spectra from which the spectral region corresponding to EHVO CIV absorption ($\approx 1150\mathring{A}$) in rest frame has been masked. The reconstruction for this region is then based on the surrounding continuum structure, and any absorption present should appear in the trough. The optimal combination of these parameter adjustments will be determined by testing a modified version of SpenderQ on confirmed EHVO candidates from variability studies by Rodriguez Hidalgo et al. (2022) [20], whose absorption is known, using different combinations of adjustments. The initial integration of SpenderQ is described in the following section.

5.7 Results

I ran SpenderQ independently for the ten spectral observations of five confirmed NV cases and the three J2318 observations described in Section 4, using an implementation of SpenderQ adjusted to fit our data (Figure 5). Each SDSS spectrum is median-normalized in a rest-frame continuum window (1445–1455$\mathring{A}$), resampled to the SpenderQ grid, and reconstructed using iterative Lyα-forest masking following the procedure used by Hahn et al. [7]. The output is rebinned to the observed wavelength grid, rescaled to match the observed flux in the normalization window, and used to form continuum-normalized spectra, $T(\lambda) = f_{\rm obs}/f_{\rm recon}$ (Figure 4).

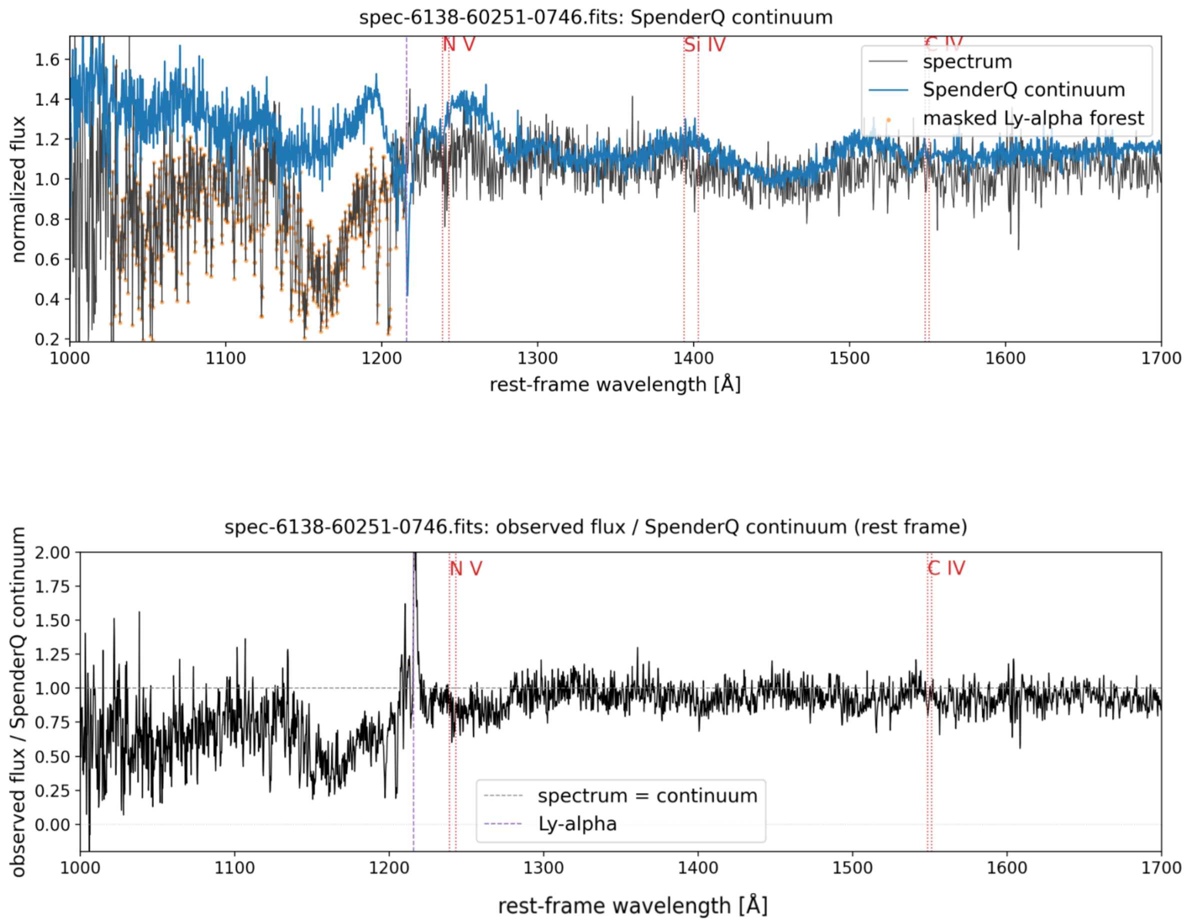

FIG. 4. SpenderQ continuum reconstruction and normalization for J2318 ($z = 2.678$; MJD 60251). (a) Observed spectrum (black) and SpenderQ unabsorbed continuum (blue) in rest frame. Orange points mark pixels masked as Lyα-forest absorption. Vertical lines indicate Lyα (purple dashed) and NV, SiIV, and CIV (red dotted) emission lines. (b) Transmission spectrum $T(\lambda) = f_{\rm obs}/f_{\rm recon}$ in the rest frame. The dashed line at 1 marks the continuum level.

Across 10 spectra epochs from the five NV quasars described in Section 3 ($z \approx 2.7$–3.6) and three epochs of J2318 ($z = 2.678$), SpenderQ largely reproduces the broad quasar continuum shape and emission lines and traces an upper shape through the Lyα forest (Figure 5). In some cases, the reconstructed continuum is underestimated , for example $J103800.50$ ($z = 3.623$; MJD 56661, MJD 52428) Figure 5).

Multi-epoch ratios of reconstructed quasar continua are consistent with dividing to 1 at 1450 Å(median ≈ 1.000 as shown in Figure 8), indicating a consistent intrinsic quasar continuum modeled for two independent observation epochs for $J230820.39$ ($z = 3.623$; MJD 56246, MJD 56980). The

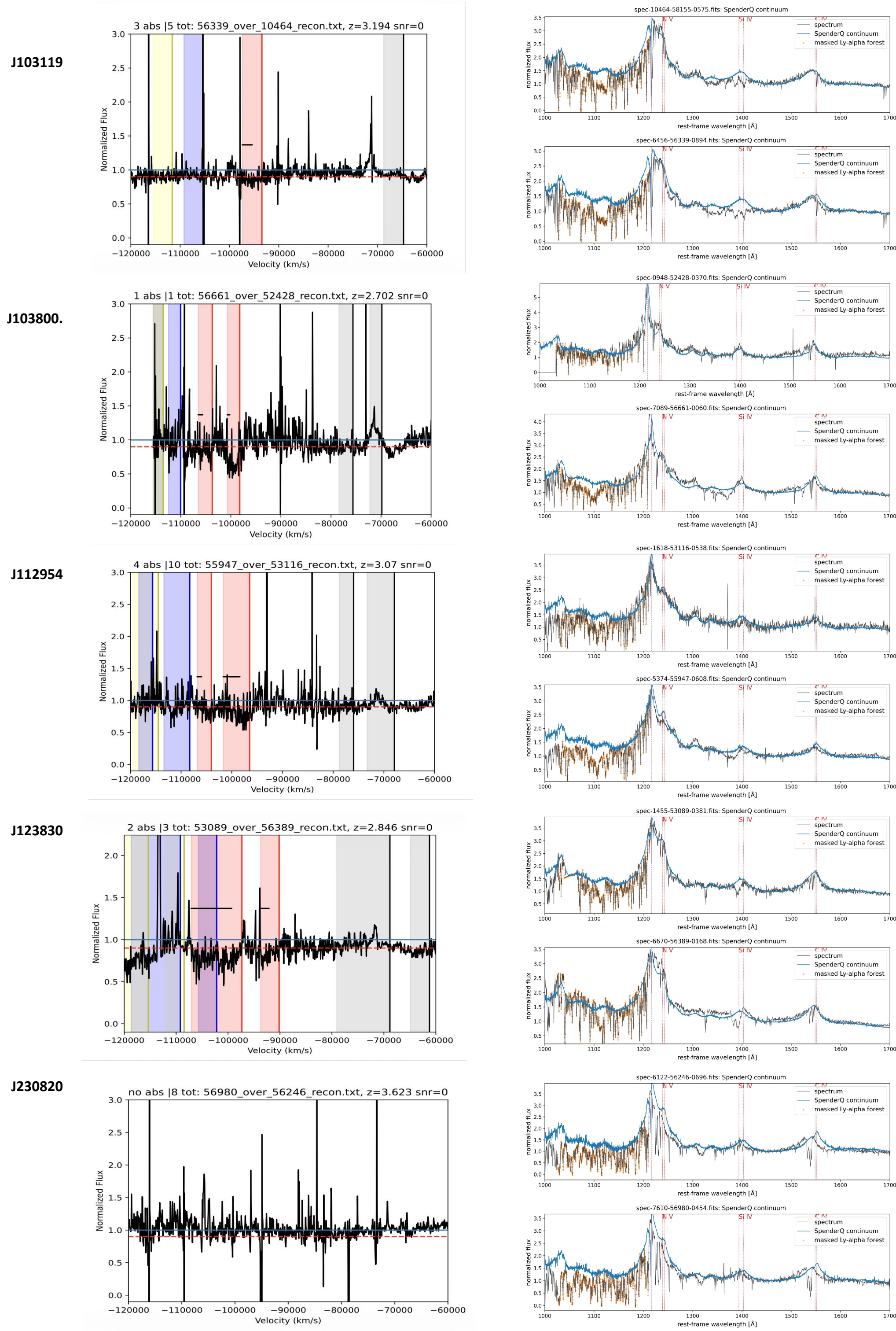

FIG. 5. (Right) SpenderQ continuum reconstructions (blue) for quasars with two observation epochs each from the NV variability sample (five quasars from SDSS DR16, $z \approx 2.7$–3.6). Orange points mark masked Ly-α absorption lines. Vertical lines mark Ly-α emission (purple) and NV, SiIV, and CIV (red) emission lines in the quasar rest frame. Rows are labeled by quasar name, with two spectra corresponding to each epoch of observation. For each quasar, the ratio of the spectra (left) after normalization via their respective reconstructions is flagged for absorption using the automated absorption detection module. Absorption is automatically marked by the horizontal black line, indicating flux $> 10\%$ below the continuum and wider than 1000 km/s in velocity space.

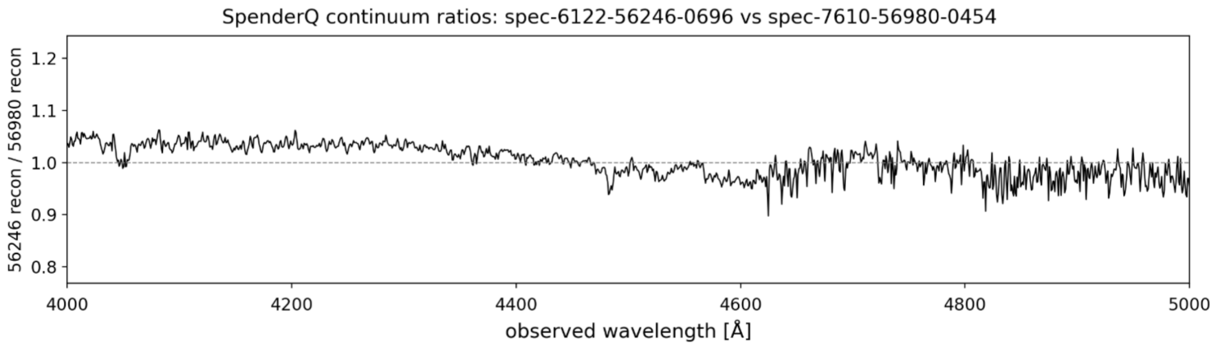

FIG. 6. Multi-epoch stability of SpenderQ continuum reconstructions for $J230820.39$ ($z = 3.623$; MJD 56246, MJD 56980), showing the ratio of reconstructed continua between two epochs in the observed frame. Ratios remain near 1 across the wavelength range of interest, demonstrating that the model assigns a consistent intrinsic continuum to repeat observations of the same quasar. The median ratio is ≈1.0 (0.9–1.05) across the normalization window $\approx 4000 - 5000$ Å, showing that the reconstructions are stable where data quality is good, supporting the use of these reconstructions for continuum-normalization.

similarity of these reconstructions across different epochs provides initial evidence that normalizing spectra via the ML-reconstructed unabsorbed continua is a possible approach for future searches. Despite the slight underestimation in some cases, reconstructed continua are consistently similar across multi-epoch observations in the 1445–1455 Åwindow, with a mean range of $T(\lambda) = 0.95 - 1.05$ across the five NV cases. Since autoencoder reconstructions are mutually independent [11], similar reconstructions of the underlying quasar continuum at different epochs validates the intended normalization point. Adjustments to improve the accuracy of the reconstructions (Section 5.2) will be explored in future work. The reconstructed unabsorbed continua for each spectrum were then divided as described in Section 3, to expose variability in flux across observations compared to the unabsorbed continua. All ratios for every quasar were produced simultaneously and passed through the absorption detection module. As shown in (Figure 5), absorption is detected in the expected velocity space for all 5 quasars from the NV dataset.

Identification of NV absorption within the Lyα-forest across four of five cases provides evidence that dividing spectra by their SpenderQ-reconstructed continua effectively surpresses Lyα lines to isolate variable intrinsic absorption. To validate detection of NV within the Lyα region, absorption identified via the SpenderQ continuum-normalized spectral ratios will be compared against ratios of these spectra normalized by the power law via the pipeline introduced in Section 3 in future work.

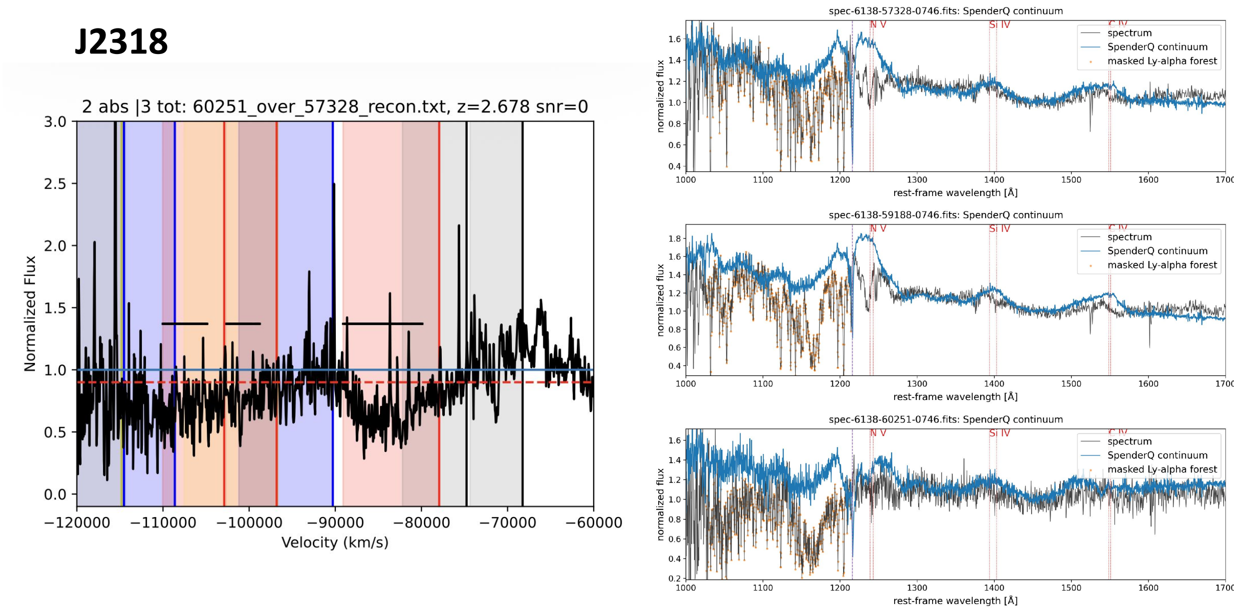

FIG. 7. (Right) SpenderQ continuum reconstructions (blue) for quasar J2318 ($z = 2.678$, MJD 57328, MJD 59188, MJD 60251). Orange points mark masked Ly-α absorption lines. Vertical lines mark Ly-α emission (purple) and NV, SiIV, and CIV (red) emission lines in the quasar rest frame. The ratio between two observations (MJD 60251 / MJD 57328) (left) after normalization via their respective reconstructions is flagged for absorption using the automated absorption detection module. Absorption is automatically marked by the horizontal black line, indicating flux $> 10\%$ below the continuum and wider than 1000 km/s in velocity space.

J123830.71

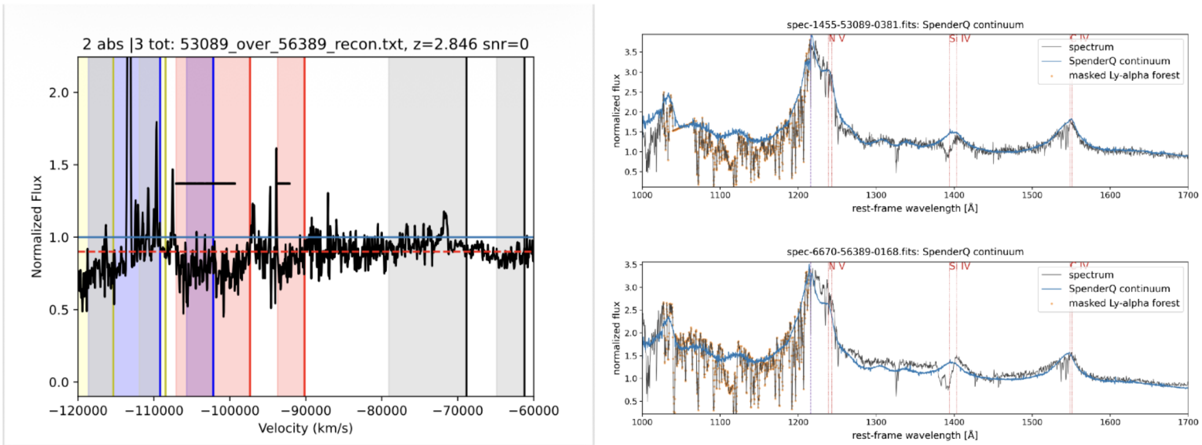

FIG. 8. Reconstructions (right) of the unabsorbed quasar continua for two observations of quasar J123830.71 (z = 2.846; MJD 53089, MJD 56389). The original spectra (black) were individually divided by their respective unabsorbed continua reconstructions (blue), effectively normalizing them to the model's estimated continua. These continuum-normalized spectra were then passed through our automated absorption detection module. Absorption is detected by the program in the velocity ranges -92,000 $km/s < v <$ -95,000 km/s and -98,000 $km/s < v <$ -109,000 km/s,

6. DISCUSSION: BROADER SIGNIFICANCE

Our pipeline is intended to scale to larger datasets of quasar spectra through its modular architecture. Reanalyzing all SDSS-V quasar spectra to carry out the first systematic search for EHVOs at $v > 0.2c$, using the methods tested in this paper, could dramatically increase the most extreme cases of EHVOs. A substantially larger confirmed EHVO sample could provide the statistical evidence needed to test theoretical models of AGN-galaxy feedback mechanisms.

7. CONCLUSIONS AND FUTURE WORK

Confirming EHVO absorption in the $v > 0.2c$ domain is extremely challenging, since we cannot rely on conventional spectral analysis methods such as power law estimation and template fitting procedures and visual inspection from individual observations is unmanageable given the number of quasars detected by SDSS. Multi-epoch spectra ratios and ML-reconstructed quasar continua show promising initial results. Moving forward, one potential path is to combine several detection approaches rather than depending on any single method. Running the ratio method, autoencoder-reconstructed continua, and a fitting method such as PCA through separate modules, then checking whether the same absorption features are flagged consistently across all three, could increase detection confidence. Since the pipeline is already modular, this kind of verification using multiple methods is a possible next step. Building a statistically reliable catalog of the fastest known quasar outflows will require unconventional methods, and the framework developed here provides a foundation to work from. While the main stages of the pipeline have been successfully validated for one known EHVO with $v > 0.2c$, further progress is needed to achieve full automation and statistical verification. First, parameters that guide reconstruction in the network layers–specifically latent dimensionality, deviation, and masking suspected CIV regions–will be adjusted (as described in Section 5.2) to determine the combination of parameters that produces the most accurate continuum reconstruction. Accuracy of the reconstructed continua will be evaluated based on how consistently and closely the model fits common features across multi-epoch observations (as described in Section 5.7). Second, the pipeline will be tested against a larger sample of SDSS quasars to search for CIV absorption at $v > 0.3c$, as well as a sample of quasars without EHVOs to validate against false positives. Two open questions will shape the direction of future work. First, how well does the autoencoder reconstruct continua for quasars with different emission properties and signal-to-noise conditions? Specifically, at what point does reconstruction quality decrease enough to introduce false positives or miss real absorption? Comparison across a larger set of confirmed cases will help answer this. Second, once the pipeline is scaled to the full SDSS-V catalog, will the confirmed sample of $v > 0.2c$ EHVOs be large enough to begin testing theoretical predictions about the relationship between outflow velocity and black hole mass with confirmed data? The answers to these questions will determine whether the framework developed here can move from a prototype to a tool that significantly contributes to the broader picture of AGN feedback in the early universe.

GLOSSARY OF TERMS

1. EHVO

EHVOs are energetic ionized gas generally considered to launch from quasar accretion disks at speeds exceeding 10% of the speed of light ($v > 0.1c$) [21]. This paper focuses on a new velocity range of EHVOs exceeding 20% the speed of light ($0.2c$). Kinetic power scales with velocity cubed: an outflow moving at 0.2c carries roughly one to two and a half orders of magnitude more energy than one moving at the high-velocity threshold of 5,000–10,000 km s^{-1}, assuming similar distances from the central source [19]. This massive kinetic power output places EHVOs among the most powerful energy feedback mechanisms in the universe [18]. Over 150 EHVOs have been confirmed to date by the quasar research group at the University of Washington Bothell [19] (Candelaria-Stoner 2026, in preparation), establishing them as a common rather than anomalous feature of quasar activity. However, the fastest EHVOs ($v > 0.2c$) are severely under-detected due to their blueshift into the Ly-α forest region, which obscures their identifying spectral absorption features. This paper introduces new computational tools to extend the detectable velocity range of EHVO surveys beyond what most previous methods allowed.

2. Ly-α Forest (Lyman-Alpha Forest)

The Lyman-Alpha forest is the dense collection of narrow hydrogen absorption lines observed in the spectra of distant quasars, caused by the many intervening hydrogen gas clouds in the intergalactic medium (IGM) at a variety of redshifts. Each cloud produces an absorption line at a wavelength slightly different from every other cloud, producing a characteristic "forest" of lines blueshifted from the quasar's own emission lines [17]. Unlike the intrinsic absorption features of quasar outflows, Ly-α forest lines are non-variable on human timescales because the intervening gas clouds do not change significantly over the span of years between observations. They are effectively constant in contrast to the highly variable EHVO winds [1]. This non-variability is the physical foundation of the spectral ratio method used in this paper to isolate EHVO signals. The forest is especially problematic for detecting the fastest EHVOs since outflow velocities above $0.2c$ shift ion absorption identifiers (such as CIV) directly into this crowded spectral region [17].

3. CIV

CIV refers to triply ionized carbon (a carbon atom that has lost six of its electrons) and is one of the most prominent ultraviolet absorption lines used to identify quasar outflows. In a quasar spectrum at rest frame, CIV absorption appears as a characteristic doublet near 1550 Å. In the context of EHVO detection, the CIV absorption doublet is used as the primary identifier of an outflow. The further the absorption feature is blueshifted to shorter wavelengths in the observed spectrum, the higher the velocity of the outflowing gas which is moving in the direction toward Earth from the reference frame of the quasar. CIV is typically the strongest and most reliable identifying feature, and its presence confirms an EHVO candidate [19].

4. Autoencoder (SpenderQ)

An autoencoder is a type of neural network that learns to compress input data into a lower-dimensional numerical representation (encoding), and then reconstruct the original data from that compressed form (decoding) [6]. SpenderQ is a specific autoencoder model applied to quasar spectra [7]. It takes the full flux array of a spectrum as input and uses the layers of its network to convert the essential spectral information into approximately 10 parameters, for example the strength of CIV and NV emission lines and the slope of the continuum [11]. The critical advantage of SpenderQ for our research is that it is redshift-invariant: it can be trained once on a set of simulated or known spectra and then applied to quasar observations at any redshift [7]. It does not require retraining using separate data, which eliminates the need for mock data or splitting the limited observational data that is currently accessible. The goal of integrating SpenderQ into our detection pipeline is to reconstruct the unabsorbed quasar continuum, then to divide multi-epoch observations by their corresponding unabsorbed continua. This way, EHVO absorption features will ideally be made visible in cases where the ratio method is insufficient.

5. SDSS (Sloan Digital Sky Survey)

The Sloan Digital Sky Survey is a large-scale spectroscopic and photometric sky survey that has produced spectra for millions of astronomical objects, including quasars, using a dedicated 2.5-meter telescope at Apache Point Observatory in New Mexico. SDSS data provides the multi-epoch spectral observations that are essential for the pipeline developed in this paper. Since the most recent SDSS-V survey has collected multi-epoch observations of over 300,000 quasars, it enables the ratio-based method described in Section 3. Future SDSS surveys such as The After Sloan-V (AS5) [**?**] will likely allow for an even larger set of potential EHVO candidates at the higher velocity range targeted by this research.

6. AGN (Active Galactic Nucleus)

An active galactic nucleus is the extremely luminous, compact central region of a galaxy in which a supermassive black hole is actively accreting surrounding matter. This accretion disk releases massive amounts of energy across the electromagnetic spectrum. Quasars are the most luminous type of AGN, often outshining their entire host galaxies by several orders of magnitude [10]. The energetic outflows produced by AGN [18], including EHVOs, are thought to play a central role in regulating galaxy growth through a process called AGN feedback, in which the energy deposited by the AGN regulates star formation in the surrounding galaxy [12] /citedimatteo2025. Understanding AGN and their outflows is relevant to black hole physics as well as the broader question of how the galaxies in our universe formed from the transport of energy through these systems [4].

[1] Boisse, P., Bergeron, J., Prochaska, J. X., Péroux, C., and York, D. G. (2015). Time variations of narrow absorption lines in high resolution quasar spectra. *Astronomy & Astrophysics*, 581:A109.

[2] Busca, N., Rich, J., Bautista, J., Cuceu, A., Font-Ribera, A., Guy, J., Herrera-Alcantar, H. K., Stermer, J., Balland, C., Aguilar, J., Ahlen, S., Bianchi, D., Brooks, D., Claybaugh, T., de la Macorra, A., Doel, P., Ferraro, S., Forero-Romero, J. E., Gaztañaga, E., Gordon, C., Gutierrez, G., Ishak, M., Kehoe, R., Kirkby, D., Kremin, A., Landriau, M., Guillou, L. L., Magneville, C., Martini, P., Miquel, R., Nadathur, S., Palanque-Delabrouille, N., Prada, F., Pérez-Ràfols, I., Ravoux, C., Rossi, G., Sanchez, E., Schlegel, D., Seo, H., Silber, J., Sprayberry, D., Tarlé, G., Weaver, B. A., Zhou, R., and Zou, H. (2025). The effects of continuum fitting on lyman-α forest correlations.

[3] Busca, N. G., Delubac, T., Rich, J., Bailey, S., Font-Ribera, A., Kirkby, D., Le Goff, J.-M., Margala, D., Slosar, A., Aubourg, , and et al. (2013). Baryon acoustic oscillations in the lyα forest of boss quasars. *Astronomy & Astrophysics*, 552:A96.

[4] Di Matteo, T., Springel, V., and Hernquist, L. (2005). Energy input from quasars regulates the growth and activity of black holes and their host galaxies. *Nature (London)*, 433(7026):604–607.

[5] Gebhardt, K., Bender, R., Bower, G., Dressler, A., Faber, S. M., Filippenko, A. V., and Tremaine, S. (2000). A relationship between nuclear black hole mass and galaxy velocity dispersion. *The Astrophysical Journal Letters*, 539(1):L13–L16.

[6] Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep Learning*, chapter 14. MIT Press.

[7] Hahn, C., Gontcho A Gontcho, S., Melchior, P., Herrera-Alcantar, H. K., Aguilar, J. N., Ahlen, S., Bianchi, D., Brooks, D., Claybaugh, T., de la Macorra, A., Dey, A., Doel, P., Forero-Romero, J. E., Gutierrez, G., Ishak, M., Juneau, S., Kirkby, D., Kisner, T., Kremin, A., Lambert, A., Landriau, M., Le Guillou, L. ., and Zou, H. (2025). Reconstructing quasar spectra and measuring the lyman-alpha forest with spenderq.

[8] Hamann, F., Chartas, G., Reeves, J., and Nardini, E. (2018). Does the X-ray outflow quasar PDS 456 have a UV outflow at 0.3c? , 476(1):943–953.

[9] Kollmeier, J. et al. (2017). Sdss-v: Pioneering panoptic spectroscopy. *Astrophysics of Galaxies*.

[10] Longair, M. (2003). *The Quasar Enigma*. University of Chicago Press.

[11] Melchior, P., Liang, Y., Hahn, C., and Goulding, A. (2023). Autoencoding galaxy spectra. i. architecture. *The Astronomical Journal*, 166(2):74.

[12] Morganti, R. (2017). The many routes to agn feedback. *Frontiers in Astronomy and Space Sciences*, 4:42.

[13] NASA (2024). Hubble quasars. `https://nasa.gov`. Accessed: 2026-05-27.

[14] Peterson, B. M. (1997). *An Introduction to Active Galactic Nuclei*. Cambridge University Press.

[15] Pierce, E., Vong, A., Wang, A., DeFrancisco, C., Rijal, A., Romo Pérez, T., Charles, M., García Naranjo, W., Fulda, R., Powell, V., and Sanabria, L. (2023). Systematic analysis of the quasar properties and variability of extremely high velocity outflow quasars in the sloan digital sky survey. *The Crow: Campus Research and Observational Writings from the University of Washington Bothell*, 8.

[16] Rauch, M. (1998). The lyman alpha forest in the spectra of quasistellar objects. *Annual Review of Astronomy and Astrophysics*, 36:267–316.

[17] Rauch, M. (2001). Lyman alpha forest. In *Encyclopedia of Astronomy and Astrophysics*. Nature Publishing Group and Institute of Physics Publishing.

[18] Rodríguez Hidalgo, P., Choi, H., Hall, P. B., Leighly, K. M., Flores, L., Charles, M. M., DeFrancesco, C., Hlavacek-Larrondo, J., and Perreault-Levasseur, L. (2025). Massive Extremely High-velocity Outflow in the Quasar J164653.72+243942.2. *Astrophys. J.*, 990(2):152.

[19] Rodriguez Hidalgo, P. et al. (2020). Survey of extremely high-velocity outflows in sloan digital sky survey quasars. *The Astrophysical Journal*, 896:151.

[20] Rodriguez Hidalgo, P. et al. (2022). Connection between emission and absorption outflows through the study of quasars with extremely high velocity outflows. *The Astrophysical Journal Letters*, 939:L24.

[21] Rodriguez Hidalgo, P., Hamann, F., and Hall, P. (2011). The extremely high velocity outflow in quasar PG0935+417. *Monthly Notices of the Royal Astronomical Society*, 411(1):247–259.

[22] Seaton, L. M., Hall, P. B., Flores, L., Hidalgo, P. R., Veltri, M., Zhu, Z., Serna, J., Brandt, W. N., Anderson, S., Assef, R. J., and et al. (2026). A new member of the fast and furious family: A relativistic and time-variable uv outflow in a luminous quasar. *The Astrophysical Journal*, 1004(1):49.